RESOLUTION MEPC.261(68)
(adopted on 15 May 2015)
AMENDMENTS TO THE 2014 GUIDELINES ON SURVEY AND CERTIFICATION
OF THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)
(RESOLUTION MEPC.254(67))MEPC 68/21/Add.1
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(RESOLUTION MEPC.254(67))

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,

RECALLING ALSO that, at its sixty-second session, it adopted, by resolution MEPC.203(62), Amendments to the Annex of the Protocol of 1997 to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING that the aforementioned amendments to MARPOL Annex VI entered into force on 1 January 2013,

NOTING ALSO that regulation 5 (Surveys) of MARPOL Annex VI, as amended, requires ships to which chapter 4 applies shall also be subject to survey and certification taking into account guidelines developed by the Organization,

NOTING FURTHER that, at its sixty-third session, it adopted, by resolution MEPC.214(63), 2012 Guidelines on survey and certification of the Energy Efficiency Design Index (EEDI) , which were further amended at its sixty-fifth session, by resolution MEPC.234(65),

NOTING FURTHER that, at its sixty-seventh session, it adopted, by resolution MEPC.254(67), 2014 Guidelines on survey and certification of the Energy Efficiency Design Index (EEDI),

RECOGNIZING that the amendments to MARPOL Annex VI require the adoption of relevant guidelines for the smooth and uniform implementation of the regulations and to provide sufficient lead time for industry to prepare,

Index (EEDI),

RECOGNIZING that the amendments to MARPOL Annex VI require the adoption of relevant guidelines for the smooth and uniform implementation of the regulations and to provide sufficient lead time for industry to prepare,

HAVING CONSIDERED, at its sixty-eighth session, draft amendments to the 2014 Guidelines on survey and certification of the Energy Efficiency Design Index (EEDI),

1 ADOPTS amendments to the 2014 Guidelines on survey and ce rtification of the Energy Efficiency Design Index (EEDI), as set out in the annex to the present resolution;

2 INVITES Administrations to take the aforementioned amendments into account when developing and enacting national laws which give force to and implement provisions set forth in regulation 5 of MARPOL Annex VI, as amended;

3 ENDORSES the use of ISO standard 15016:2105 for ships for which the sea trial is conducted on or after 1 September 2015 and encourages the application of the standard prior to that date;

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prior to that date;

4 REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the amendments to the attention of shipowners, ship operators shipbuilders, ship designers and any other interested groups;

5 AGREES to keep these guidelines, as amended, under review in light of the experience gained with their application.

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AMENDMENTS TO THE 2014 GUIDELINES ON SURVEY AND CERTIFICATION
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1 Paragraphs 4.3.5 and 4.3.6 are replaced with the following:

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(RESOLUTION MEPC.254(67))

1 Paragraphs 4.3.5 and 4.3.6 are replaced with the following:

"4.3.5 Sea conditions should be measured in accordance with ITTC Recommended Procedure 7.5-04-01-01.1 Speed and Power Trials Part 1; 2014 or ISO 15016:2015.

4.3.6 Ship speed should be measured in accordance with ITTC Recommended Procedure 7.5-04-01-01.1 Speed and Power Trials Part 1 ; 2014 or ISO 15016:2015, and at more than two points of which range includes the power of the main engine as specified in paragraph 2.5 of the EEDI Calculation Guidelines."

2 Paragraphs 4.3.8 and 4.3.9 are replaced with the following:

"4.3.8 The submitter should develop power curves based on the measured ship speed and the measured output of the main engine at sea trial. For the development of the power curves, the submitter should calibrate the measured ship speed, if necessary, by taking into account the effects of wind, current, waves, shallow water, displacement, water temperature and water density in accordance with ITTC

of the power curves, the submitter should calibrate the measured ship speed, if necessary, by taking into account the effects of wind, current, waves, shallow water, displacement, water temperature and water density in accordance with ITTC Recommended Procedure 7.5-04-01-01.2 Speed and Power Trials Part 2 ; 2014 or ISO 15016:2015. Upon agreement with the shipowner, the submitter should submit a report on the speed trials including details of the power curve development to the verifier for verification.

4.3.9 The submitter should compare the power curves obtained as a result of the sea trial and the estimated power curves at the design stage. In case differences are observed, the attained EEDI should be recalculated, as necessary, in accordance with the following:

.1 for ships for which sea trial is conducted under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines : the attained EEDI should be recalculated using the measured ship speed at sea trial at the power of the main engine as specified in paragraph 2.5 of the EEDI Calculation Guidelines; and

.2 for ships for which sea trial cannot be conducte d under the

speed at sea trial at the power of the main engine as specified in paragraph 2.5 of the EEDI Calculation Guidelines; and

.2 for ships for which sea trial cannot be conducte d under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines: if the measured ship speed at the power of the main engine as specified in paragraph 2.5 of the EEDI Calculation Guidelines at the sea trial conditions is different from the expected ship speed on the power curve at the corresponding condition, the shipbuilder should recalculate the attained EEDI by adjusting ship speed under the condition as specified in paragraph 2.2 of the EEDI Calculation Guidelines by an appropriate correction method that is agreed by the verifier.

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An example of scheme of conversion from trial condition to EEDI condition at EEDI power is given as follows:

An example of scheme of conversion from trial condition to EEDI condition at EEDI power is given as follows:

Vref is obtained from the results of the sea trials at trial condition using the speed-power curves predicted by the tank tests. The tank tests shall be carried out at both draughts: trial condition corresponding to that of the S/P trials and EEDI condition. For trial conditions the power ratio P between model test prediction and sea trial result is calculated for constant ship speed. Ship speed from model test prediction for EEDI condition at EEDI power multiplied with P is Vref.
STrial
PTrial
P
P
P
,
,

where:
PTrialP ,
: power at trial condition predicted by the tank tests
STrialP ,
: power at trial condition obtained by the S/P trials
P
: power ratio

Figure 2 shows an example of scheme of the conversion to derive the resulting ship speed at EEDI condition (
refV) at EEDI power.

Figure 2: An example of scheme of conversion from trial condition to EEDI condition at EEDI power

Figure 2 shows an example of scheme of the conversion to derive the resulting ship speed at EEDI condition (
refV) at EEDI power.

Figure 2: An example of scheme of conversion from trial condition to EEDI condition at EEDI power

Note: Further consideration would be necessary for speed adjustment methodology in paragraph 4.3.9.2 of these guidelines. One of the concerns relates to a possible situation where the power curve for sea trial condition is estimated in an excessively conservative manner (i.e. power curve is shifted in a leftward direction) with the intention to get an upward adjustment of the ship speed by making the measured ship speed at sea trial easily exceed the lower-estimated speed for sea trial condition at design stage."

***
MCR
NCR
EEDI power
Power
EEDI condition Trial condition
Tank test Results
Sea Trial Results
Adjusted speed by the
Results of Sea trial

refV
Ship Speed
P * EEDI power
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